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Prediction capability of the MCDA-AHP model in wildfire risk zonation of a protected area in the Southern Western Ghats

Salma | S. Nikhil | +5 authors | R. S. Ajin | Environmental Science, Geography · Environmental Sustainability · 17 January 2023

Wildfires are a common threat to the Western Ghats region in India, and many protected areas within the Ghat region have been severely battered in the past. This study aims to identify the **wildfire**... Expand

16 | Springer Nature | Save | Cite

Deep Learning for Wildfire Risk Prediction: Integrating Remote Sensing and Environmental Data

Zhengsen Xu | Jonathan Li | +8 authors | Linlin Xu | Environmental Science, Computer Science · 2 May 2024

TLDR Key challenges and limitations of current **wildfire**-risk **prediction** models are identified, and cross-disciplinary methods--such as coupling with numerical weather-**prediction** models--are incorporated to enhance the accuracy and robustness of **wildfire**-risk assessments. Expand

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LinU-Mamba: Visual Mamba U-Net with Linear Attention to Predict Wildfire Spread

Henintsoa S. Andrianarivony | M. Akhloufi | Environmental Science, Computer Science · Remote Sensing · 6 August 2025

TLDR LinU-Mamba, a model with a U-Net-based vision Mamba architecture, with light spatial attention in skip connections, and an efficient linear attention mechanism in the encoder and decoder to better capture salient fire information in the dataset is proposed. Expand

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Paying Attention to Wildfire: Using U-Net with Attention Blocks on Multimodal Data for Next Day Prediction

Jack Fitzgerald | Ethan Seefried | James E Yost | S. Pallickara | Nathaniel Blanchard

Environmental Science, Computer Science · International Conference on Multimodal... · 9 October 2023

TLDR This work created a custom U-Net architecture designed to train as efficiently as possible, while still maximizing accuracy, to facilitate quickly deploying the model when a **wildfire** is detected and demonstrates state-of-the-art performance. Expand

9 | PDF | ACM | Save | Cite

Predicting fire spread and behaviour on the fireline. Wildfire analyst pocket: A mobile app for wildland fire prediction

S. Monedero | J. Ramírez | A. Cardil | Environmental Science, Computer Science · Ecological Modelling · 24 January 2019

41 | Publisher | Save | Cite

Dengue Spread Prediction Using Time Series Model

Shushant Sharma | Shikhar Sharma | Deepesh Mishra | Environmental Science, Medicine · International Journal for Research in Applied... · 30 November 2023

TLDR This research project introduces an innovative framework that combines time series analysis and vision-based methodologies to forecast the **spread** of dengue, using Python

language and modules within the language to develop and evaluate time series forecasting models. Expand

Publisher Save Cite

From Convolution to Attention: Transformer-Based Modeling for Multi-Day Wildfire Spread Forecasting

Parul Dubey Nitin Rakesh Pushkar Dubey P. Bokoro

Environmental Science, Computer Science · IEEE Access · 2025

TLDR Results show the proposed Transformer + BDI system significantly outperforms prior CNN-based approaches, achieving an F1 score of 0.75 and reducing **prediction** MAE to 7.4 km2 across datasets, while offering interpretable and actionable **wildfire** response insights. Expand

IEEE Save Cite

Wildfire spread forecasting with Deep Learning

Nikolaos Anastasiou Spyros Kondylatos Ioannis Papoutsis

Environmental Science, Computer Science · arXiv.org · 23 May 2025

TLDR A deep learning (DL)-based framework for forecasting the final extent of burned areas, using data available at the time of ignition, using a spatio-temporal dataset that covers the Mediterranean region from 2006 to 2022 is presented. Expand

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Simulating wildfire spread based on continuous time series remote sensing images and cellular automata

Hua Zhuang Naian Liu +4 authors Rui Wang Environmental Science, Computer Science · International journal of wildland fire · 16 January 2025

TLDR An adaptive simulation method based on continuous time remote sensing images and the CA model, which effectively characterised gradual **spread** based on low ROS, serves as a powerful tool for wildland surface fire prevention and control. Expand

Publisher Save Cite

Comparative and Interpretative Analysis of CNN and Transformer Models in Predicting Wildfire Spread Using Remote Sensing Data

Yihang Zhou Ruige Kong Zhengsen Xu Linlin Xu Sibo Cheng

Environmental Science, Computer Science · Journal of Geophysical Research · 18 March 2025

TLDR UNet and Transformer-based Swin-UNet are able to focus on critical features such as “Previous Fire Mask”, “Drought”, and “Vegetation” more effectively than the other two models, while also maintaining balanced attention to the remaining features, leading to their superior performance. Expand

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